

HUNTERS POINT, MB, Canada

WMO: 711421

Lat: **53.027N**Lon: **100.935W**Elev: **256**StdP: **98.29**Time Zone: **-6.00 (NAC)**Period: **94-05**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.8	(c) -30.5	(d) -36.5	(e) 0.1	(f) -32.6	(g) -34.3	(h) 0.2	(i) -30.4	(j) 13.8	(k) -18.7	(l) 12.6	(m) -19.0	(n) 3.3	(o) 280	(p) 0.735

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.2	(c) 27.0	(d) 20.0	(e) 25.3	(f) 20.1	(g) 23.9	(h) 19.8	(i) 22.2	(j) 25.0	(k) 21.3	(l) 24.1	(m) 20.3	(n) 23.1	(o) 3.2	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	16.4	24.2	20.3	15.5	23.3	19.2	14.4	22.3	66.9	25.1	63.4	24.1	59.7	23.0	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.3	(b) 11.8	(c) 10.4	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 13.3	(b) 11.8	(c) 10.4	(d) WB	(e) -36.2	(f) 23.8	(g) 1.9	(h) 1.2	(i) -37.6	(j) 24.7	(k) -38.7	(l) 25.4	(m) -39.7	(n) 26.1	(o) -41.1	(p) 26.9	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-18.6	-14.5	-9.1	-1.2	6.5	15.8	19.5	18.4	12.7	4.3	-5.3	-13.7	
	DBStd	14.28	7.79	7.64	7.82	5.42	5.30	3.92	2.75	3.41	4.13	4.73	6.13	8.24	
	HDD10.0	4042	886	686	592	336	137	4	0	0	22	183	461	736	
	HDD18.3	6333	1144	919	851	586	368	93	20	41	172	436	711	994	
	CDD10.0	873	0	0	0	1	30	179	293	262	103	6	0	0	
	CDD18.3	123	0	0	0	0	2	18	56	44	3	0	0	0	
	CDH23.3	465	0	0	0	0	18	91	191	150	15	1	0	0	
	CDH26.7	84	0	0	0	0	6	27	26	24	2	0	0	0	
Wind	WSAvg	4.8	4.4	4.6	4.6	4.4	4.5	4.8	4.4	4.9	5.4	5.7	5.2	4.7	
Precipitation	PrecAvg	507	22	15	19	27	45	79	87	69	58	41	25	21	
	PrecMax	703	40	29	44	48	105	136	153	170	110	120	61	46	
	PrecMin	400	9	4	4	6	11	52	50	28	16	5	9	10	
	PrecStd	76	8	6	9	13	23	20	26	35	26	22	11	8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	4.6	9.4	16.2	25.6	30.1	29.3	30.0	25.3	16.4	7.6	6.2	
		MCWB	3.5	1.8	5.6	10.1	13.8	19.2	20.3	19.8	16.8	12.2	3.8	2.7	
	2%	DB	1.5	1.4	5.3	9.0	19.1	25.4	26.9	26.3	22.0	13.7	4.7	2.0	
		MCWB	0.2	0.5	2.6	5.3	13.7	19.9	20.5	20.9	16.3	9.9	2.5	0.3	
	5%	DB	-2.9	-0.5	3.3	6.2	16.0	23.1	25.3	24.8	19.8	11.8	3.2	-0.9	
		MCWB	-3.8	-1.4	1.6	3.6	12.8	18.4	20.2	20.5	16.0	9.2	1.8	-1.9	
	10%	DB	-7.2	-2.8	1.6	4.6	13.9	21.2	24.0	23.5	18.0	10.2	2.1	-3.3	
		MCWB	-7.6	-3.5	0.7	2.7	11.0	17.4	20.1	19.9	14.9	8.4	1.1	-4.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	2.3	5.5	8.9	16.4	22.7	23.7	23.4	18.8	13.0	4.3	2.9
			MCDB	6.1	3.7	9.1	15.7	23.1	25.5	26.0	25.7	21.5	16.4	6.7	6.2
2%		WB	0.3	0.6	2.9	5.3	14.2	20.9	22.4	22.1	17.5	10.5	2.9	0.3	
		MCDB	1.2	1.4	4.8	8.2	18.0	24.4	24.9	25.0	20.5	12.7	4.4	1.7	
5%		WB	-3.8	-1.4	1.7	3.8	12.7	19.2	21.5	21.2	16.3	9.5	2.0	-1.9	
		MCDB	-2.9	-0.4	3.1	6.0	15.7	22.3	24.0	24.2	19.3	11.3	3.2	-0.9	
10%		WB	-7.6	-3.5	0.7	2.8	11.1	17.9	20.5	20.2	15.1	8.5	1.2	-3.9	
		MCDB	-7.3	-2.8	1.7	4.5	13.7	20.7	23.0	23.1	17.7	10.2	2.1	-3.3	
Mean Daily Temperature Range		5% DB	MDBR	8.5	9.9	11.3	9.0	7.4	6.5	7.2	6.8	6.3	5.1	5.8	7.8
			MCDBR	13.0	10.7	12.5	11.1	10.2	10.0	9.5	9.0	9.7	7.5	6.6	10.4
	MCWBR		11.7	9.7	10.3	7.9	5.7	5.7	5.5	5.4	5.7	5.0	5.2	9.1	
	5% WB	MCDBR	12.7	10.4	11.7	10.2	9.0	7.6	7.5	7.6	8.1	6.8	6.2	10.3	
		MCWBR	11.5	9.5	9.8	7.5	5.3	5.0	5.2	5.2	5.5	4.7	4.9	9.0	
Clear-Sky Solar Irradiance	taub		0.252	0.257	0.282	0.347	0.357	0.386	0.390	0.373	0.332	0.294	0.255	0.233	
	taud		2.265	2.311	2.383	2.379	2.412	2.363	2.392	2.434	2.531	2.576	2.429	2.306	
	Ebn at Noon		744	869	916	887	893	866	854	851	853	825	754	710	
	Edh at Noon		60	79	92	107	109	116	111	100	81	62	51	48	
All-Sky Solar Radiation	RadAvg		1.05	2.21	3.71	4.94	5.51	5.59	5.81	4.76	3.37	1.89	0.91	0.71	
	RadStd		0.09	0.13	0.25	0.41	0.51	0.44	0.33	0.40	0.35	0.25	0.13	0.09	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	+1.69	N/A	-0.38	-0.43	N/A	N/A	N/A	N/A

Nomenclature: See separate page